

Syllabus Details

Syllabus ID:	12841
Syllabus Name:	Operating System_Hệ điều hành
Course Name English:	Operating System
Subject Code:	OSG202
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study'
Pre-Requisite:	
Description:	At the end of this course, students will be able to perceive: 1) Background knowledge: (ABET e) -The role of operating system, - important OS concepts, - the mechanism of operating system, and - main problems of Operating system. 2) Practice skills: (ABET k) - Using basic shell command in Linux system fluently. - Fundamental of shell, C language on Linux. - Main problems of Operating system through some simulative exercises. 3) Information searching, reading and selection skills (ABET i) 4) Working in group, documenting and presenation skills (ABET g)
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU LMS https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	- Internet - Linux operating system server and/ or desktop - MARIE simulator - AI tools (ChatGPT/Copilot/Gamma...) and other simulation software if deemed appropriate.
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	437/QĐ-ĐHFPT dated 04/25/2025
IsApproved:	True
Note:	
MinAvgMarkToPass:	0
IsActive:	True
ApprovedDate:	4/25/2025

5 material(s)

MaterialDescription	Author	Publisher	PublishedDate	Edition	ISBN	IsMainMaterial	IsHardCopy	IsOnline	Note
Modern Operating Systems (4th edition or 3rd or 2nd)	Andrew S. Tanenbaum	Pearson	2014 (4th)	4th or 3rd or 2nd	9781292061429 or 9780133591620 (4th Edition)	☒	☒	☐	
Connecting with Computer Science	Creg Anderson, David Ferro, Robert Hilbon	Thomson course Technology	2010	2nd Edition	978-1439080351	☐	☒	☐	
Understanding operating systems	Ann McIver McHoes and Ida M. Flynn	Cengage Learning	2008	8th	978-1305674257	☐	☒	☐	
http://courses.knox.edu/cs226/						☐	☐	☒	
AI Notes:https://www.https://notegpt.io/						☐	☐	☒	

11 LO(s)

CLO Name	CLO Details
CLO1	Understand the main concepts of operating system
CLO10	Understand basic C/C++ language on Linux
CLO11	Using AI tools to futher explore the main concepts of OS's components
CLO2	Understand the main concepts of process management
CLO3	Understand the main concepts of memory management
CLO4	Understand the main concepts of file systems
CLO5	Understand the main concepts of I/O systems
CLO6	Understand the main concepts of deadlock
CLO7	Using Linux operating system
CLO8	Using basic shell command in Linux system fluently
CLO9	Understand basic Shell language on Linux

[View mapping of CLOs to PLOs](#)

Download All Teacher Material			Download All Student Material							
Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Lecturer Materials	T-Download	Student's Tasks	Lecturer Tasks

1	Chapter 1: Introduction 1.1. What is an operating system	Offline	L01,L011	I	Textbook/References Slides ChatGPT()	OSG202	Textbook/References Slides ChatGPT	OSG202	Lecture chapter 1 in the textbook focus on 1.1 Use ChatGPT to explore content more extensively than in the textbook based on keywords provided by the teacher, such as: single user, multi-user, single tasking, multi-tasking	Provide guidance for students o use of ChatGPT Lecture chapter 1 in the textbool focus on 1.1
2	1.2 History of operating system	Offline	L01,L011	I	Textbook/References Slides ChatGPT		Textbook/References Slides		Lecture chapter 1 in the textbook focus on 1.2 Use ChatGPT to learn more about the history of operating systems based on the keywords provided in class	Lecture chapter 1 in the textbool focus on 1.2
3	1.3. Computer hardware review	Offline	L01	I	Textbook/References Slides		Textbook/References Slides		Lecture chapter 1 in the textbook	Lecture chapter 1 in the textbool
4	Chapter 1 Problems	Offline	L01	ITU	Textbook/References Slides		Textbook/References Slides		Lecture chapter 1 in the textbook	Lecture chapter 1 in the textbool
5	Overview of Linux Operating system Overview of Linux Command	Offline	L07, L08,L011	TU	Tutorial Labs ChatGPT		Tutorial Labs ChatGPT		Research Internet/Textbook/References at home in advance Using ChatGPT to learn basic Linux operating system commands from the textbook Do practice	Guide Practice
6	Short practice of using Linux OS	Offline	L09,L011	TU	Tutorial Labs ChatGPT		Tutorial Labs		Research Internet/Textbook/References at home in advance Using ChatGPT to learn basic Linux operating system commands from the textbook Do practice	Guide Practice
7	1.4 The Operating System Zoo 1.5 Operating System Concepts	Offline	L01,L011	IT	Textbook/References Slides AI Notes (https://www.https://notegpt.io/)		Textbook/References Slides AI Notes (https://www.https://notegpt.io/)		Lecture chapter 1 in the textbook focus on 1.4, 1.5 Use 'AI Notes' to see additional explanations about different operating systems.	Provide guidance for students o Notes(https://www.https://notegpt.io/) Lecture chapter 1 in the textbool focus on 1.4, 1.5
8	1.6 System Calls 1.7 Operating System Structure	Offline	L01,L011	IT	Textbook/References Slides AI Notes (https://www.https://notegpt.io/) ChatGPT		Textbook/References Slides		Lecture chapter 1 in the textbook focus on 1.6, 1.7 Use ChatGPT/AI Notes to learn about System Calls and the structures of the Operating System	Lecture chapter 1 in the textbool focus on 1.6, 1.7
9	Chapter 1 Problems	Offline	L01, L07	U	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
10	Guiding Lab 1	Offline	L08, L09	U	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
11	Chapter 2: Processes and Threads 2.1 Processes 2.2 Threads	Offline	L02,L011	ITU	Textbook/References Slides ChatGPT		Textbook/References Slides		Lecture chapter 2 in the textbook focus on 2.1, 2.2 Use ChatGPT to learn about Threads and Processes	Lecture chapter 2 in the textbool focus on 2.1, 2.2
12	2.3 Interprocess Communication	Offline	L02,L011	ITU	Textbook/References Slides ChatGPT		Textbook/References Slides		Lecture chapter 2 in the textbook focus on 2.3 Use ChatGPT to learn about IPC and the solutions based on the keywords in the lesson	Lecture chapter 2 in the textbool focus on 2.3
13	Chapter 2 Problems Review Lab 1	Offline	L02, L07, L08, L09	U	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
14	Guiding Lab 2	Offline	L07, L08, L09	U	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
15	Chapter 2 Problems	Offline	L02, L07, L08, L09	TU	Textbook/References Slides		Textbook/References Slides		Lecture chapter 2 in the textbook	Lecture chapter 2 in the textbool
16	2.4 Scheduling	Offline	L02,L011	TU	Textbook/References Slides MARIE Simulator ChatGPT/Copilot		Textbook/References Slides MARIE Simulator ChatGPT/Copilot		Lecture chapter 2 in the textbook focus on 2.4 Use ChatGPT to learn about topics related to FCFS, Shortest Job First, Round Robin, and Priority.	Lecture chapter 2 in the textbool focus on 2.4 Guide students on how to install use the MARIE Simulator and AI ChatGPT, Gamma, Copilot to lea simulate how these scheduling algorithms run: FCFS, Shortest J First, Round Robin, Priority. Provide sample prompts to use , tools
17	2.5 Classical IPC Problems	Offline	L02, L07, L08, L09,L011	TU	Textbook/References Slides ChatGPT		Textbook/References Slides		Lecture chapter 2 in the textbook focus on 2.5 Use ChatGPT to learn about topics related to the keywords and solutions in IPC	Lecture chapter 2 in the textbool focus on 2.5
18	Chapter 2 Problems	Offline	L07,L08, L09	TU	Textbook/References Slides		Textbook/References Slides		Learn Problems	Review Problems

19	Shell programming	Offline	L07, L08, L09	T	Tutorial Labs ChatGPT		Tutorial Labs		Do lab Use ChatGPT to learn about commands related to Shell programming	Guide Lab
20	Practice shell programming	Offline	L07, L08, L09	T	Tutorial Labs		Tutorial Labs		Do lab	Guide Lab
21	Practice shell programming (continue)	Offline	L07, L08, L09	U	Tutorial Labs		Tutorial Labs		Do lab	Guide Lab
22	Review Lab 2 Chapter 2 Problems	Offline	L03, L08	T	Textbook/References Slides				Learn Problems	Review Problems Review Lab 2
23	Chapter 3: Memory management 3.1 No Memory Abstraction	Offline	L03, L08,L011	T	Textbook/References Slides AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Slides AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 3 in the textbook focus on 3.1 Use AI Notes Q&A to answer questions related to memory management in the operating system	Guide students on how to Use A to illustrate memory manageme algorithms such as LRU and FIFO Provide sample prompts to use , tools Lecture chapter 3 in the textbool focus on 3.1
24	3.2 A Memory Abstraction: Address Spaces	Offline	L03, L08,L011	T	Textbook/References slides AI Notes (https://www.https://notegpt.io/) Cached simulator		Textbook/References slides AI Notes (https://www.https://notegpt.io/) Cached simulator		Lecture chapter 3 in the textbook focus on 3.2 Use AI Notes Q&A to answer questions related to memory management in the operating system.	Provide sample prompts to use , tools Lecture chapter 3 in the textbool focus on 3.2
25	3.3 Virtual Memory	Offline	L03, L08,L011	T	Textbook/References slides AI Notes (https://www.https://notegpt.io/) Cached simulator		Textbook/References Slides		Lecture chapter 3 in the textbook focus on 3.3 Use AI Notes Q&A to answer questions related to memory management in the operating system	Lecture chapter 3 in the textbool focus on 3.3
26	3.4 Page Replacement Algorithms	Offline	L03, L08,L011	T	Textbook/References Slides ChatGPT/Copilot/Gamma MARIE Simulator		Textbook/References Slides ChatGPT/Copilot/Gamma MARIE Simulator		Lecture chapter 3 in the textbook focus on 3.4 Use AI Notes Q&A to answer questions related to memory management in the operating system	Guide students on how to use th MARIE Simulator and AI tools: ChatGPT, Gamma, and Copilot to and simulate how the algorithm. Provide sample prompts to use , tools
27	Guiding Presentation - topic's selection - Chapter 7: Virtualization and The Cloud - Chapter 10: CASE STUDY 1: UNIX, LINUX, and ANDROID - Chapter 11: CASE STUDY 2: WINDOWS 8	Offline	L03, L08	I	Textbook/References Tutorial Labs Gamma/ AI Notes (https://www.https://notegpt.io/)		Textbook/References Tutorial Labs Gamma/AI Notes		Apply the tools that have been learned to develop presentation content	Provide a general guide and con orientation for students to choose from chapter 7,1-, and 11 Guide students on how to use G AI or presentation AI combined v Note AI as a guideline to support students
28	3.5 Design Issues for Paging Systems 3.6 Implementation Issues	Offline	L03, L08,L011	I	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Slides		Lecture chapter 3 in the textbook focus on 3.5, 3.6 Use AI Notes Q&A to answer questions related to memory management in the operating system	Provide sample prompts to use , tools
29	3.7 Segmentation	Offline	L03, L08,L011	T	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 3 in the textbook focus on 3.7 Use AI Notes Q&A to answer questions related to memory management in the operating system Use ChatGPT to learn more about the mechanism and keywords related to Segmentation and other solutions	Provide sample prompts to use , tools
30	Chapter 3 Problems	Offline	L03, L08	TU	Textbook/References		Textbook/References		Lecture chapter 3 in the textbook	Lecture chapter 3 in the textbool
31	Guiding lab 3	Offline	L03, L08	TU	Textbook/References		Textbook/References		Lecture chapter 3 in the textbook	Lecture chapter 3 in the textbool
32	Progress Test 1 and Review	Offline	L03, L08	TU	Textbook/References		Textbook/References		Lecture chapter 3 in the textbook	Lecture chapter 3 in the textbool
33	Review Problems	Offline	L03, L08	TU	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
34	Chapter 4: Files 4.1 Files 4.2 Directories	Offline	L04, L09, L010	IT	Textbook/References Slides ChatGPT		Textbook/References Slides		Lecture chapter 4 in the textbook focus on 4.1, 4.2 Use ChatGPT to learn about the concepts of Files, Directories, and their organization in operating systems.	Provide sample prompts to use , tools

35	4.3 File System Implementation 4.4 File System Management and Optimization 4.5 Example File Systems	Offline	L04, L09, L010,L011	IT	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 4 in the textbook focus on 4.3, 4.4. Use ChatGPT to further explore the contents of the lesson Use ChatGPT or NoteGPT to generate simple file system simulations, then test them in Python or Java use NoteGPT to explore how these file systems worked	Provide sample prompts to use A Lecture chapter 4 in the textbool focus on 4.3, 4.4.
36	Review Lab3	Offline	L04, L09, L010		Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
37	Guiding Lab 4	Offline	L04, L09, L010	U	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
38	Chapter 4 Problems	Offline	L04, L09, L010	U	Textbook/References Slides		Textbook/References Slides		Lecture chapter 4 in the textbook	Lecture chapter 4 in the textbool
39	C programming in Linux	Offline	L04, L09, L010	T	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
40	Guiding Lab	Offline	L04, L09, L010	T	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
41	Chapter 5: Input/ Output 5.1 Principles of I/O Hardware 5.2 Principles of I/O Software	Offline	L04, L05, L09, L010	U	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Do Lab Lecture chapter 5 in the textbook focus on 5.1, 5.2 Use ChatGPT to further explore the contents of the lesson	Provide sample prompts to use A Guide Lab
42	5.3 I/O Software Layers	Offline	L05, L010,L011	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook focus on 5.3 Use ChatGPT to further explore the contents of the lesson	Provide sample prompts to use A
43	5.4 Disks	Offline	L05, L010,L011	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook focus on 5.4 Use ChatGPT to further explore the contents of the lesson	Lecture chapter 5 in the textbool focus on 5.4
44	5.5 Clocks 5.6 User Intrefaces: Keyboard, Mouse, Monitor	Offline	L05, L010,L011	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook focus on 5.6 Use ChatGPT to further explore the contents of the lesson	Lecture chapter 5 in the textbool focus on 5.6
45	5.5 Clocks 5.6 User Intrefaces: Keyboard, Mouse, Monitor	Offline	L05, L010,L011	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook focus on 5.6 Use ChatGPT to further explore the contents of the lesson	Lecture chapter 5 in the textbool focus on 5.6
46	5.7 Thin Clients 5.8 Power Management	Offline	L05, L010,L011	TU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 5 in the textbook fFocus on 5.7, 5.8 Use ChatGPT to further explore the contents of the lesson	Lecture chapter 5 in the textbool fFocus on 5.7, 5.8
47	Review lab 4	Offline	L05, L010	TU	Tutorial Labs		Tutorial Labs		Do Lab	Guide Lab
48	Chapter 5 Problems	Offline	L05, L010	TU	Textbook/References Slides		Textbook/References Slides		Lecture chapter 5 in the textbook	Lecture chapter 5 in the textbool
49	Chapter 6: Deadlock 6.1 Resources 6.2 Introduction to Deadlocks 6.3 The Ostrich Algorithm	Offline	L06, L09, L010,L011	IT	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 6 in the textbook focus on 6.1, 6.2, 6.3 Use ChatGPT to further explore the contents of the lesson Use ChatGPT/Copilot to generate scripts that describe deadlock, and then use NoteGPT/ChatGPT/Copilot to explane these scripts.	Provide sample prompts to use , tools Lecture chapter 6 in the textbool focus on 6.1, 6.2, 6.3
50	6.4 Deadlock Detection and Recovery 6.5 Deadlock Avoidance	Offline	L06, L09, L010,L011		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 6 in the textbook focus on 6.4, 6.5 Use ChatGPT to further explore the contents of the lesson	Provide sample prompts to use , tools Lecture chapter 6 in the textbool focus on 6.4, 6.5
51	6.6 Deadlock Prevention 6.7 Other Issues	Offline	L06, L09, L010,L011	IT	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 6 in the textbook focus on 6.6, 6.7 Use ChatGPT to further explore the contents of the lesson	Provide sample prompts to use , tools Lecture chapter 6 in the textbool focus on 6.6, 6.7
52	Chapter 6 Problems Guiding Problems	Offline	L06, L09, L010,L011	ITU	Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Textbook/References Tutorial Labs AI Notes (https://www.https://notegpt.io/) ChatGPT/Copilot		Lecture chapter 6 in the textbook Use ChatGPT to further explore the contents of the lesson	Lecture chapter 6 in the textbool
53	Progress Test 2 and Review	Offline	L06, L09, L010	TU						

54	Review Problems	Offline	L06, L09, L010	TU						
55	Chapter 7, 10, 11 Presentation	Offline								
56	Chapter 7, 10, 11 Presentation	Offline								
57	Chapter 7, 10, 11 Presentation	Offline								
58	Chapter 7, 10, 11 Presentation	Offline								
59	Chapter 7, 10, 11 Presentation	Offline								
60	Discussion of current research areas. Review Course	Offline								

0 Constructive question(s)

4 assessment(s)

Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
Lab	on-going	4	20.0%	>0	in lab session	L07, L08, L09, L010	practical exercises	N/A	technical aspect of studied topics	guided by instructor	
Presentation	on-going	1	20.0%	>0	~20'/group	L01, L02, L03, L04, L05, L06,L011	use Presentation Rubic sheet	N/A	a seleected topic	prepare at home, working in group, present in class hours	
Progress test	on-going	2	20.0%	>0	10'-30'	L01, L02, L03, L04, L05, L06	Multiple choicesMarked by Computer or a suitable format	10-30 (At least 10*n questions where n is the number of Chapters covered in the test.)	Current chapter(s) Each Prograss Test covers at least 01 Chapter and at most 03 Chapters, and overall they must cover all learned Chapters of the course.	in class/remote, by instructor	Instruction and shedules for Progress tests must be presented in the Course Implementation Plan approved by director of the campus. Progress test must be taken right after the last lectures of required material. Instructor has resposibility to review the test for students after graded.
Final exam	final exam	1	40.0%	4	60'	L01, L02, L03, L04, L05, L06	Multiple choicesMarked by Computer	50	concepts, algorithms; All chapters from 1 to 6; more than 70% new questions (for the current semester);	by exam board,	